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Short communication

Water entry of flexible wedges: Some issues on the FSI phenomena

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ABSTRACT

Deformable bodies entering the water can face unexpected fluid–structure interaction (FSI) phenomena introduced by the mutual interaction between the fluid motion and the structural deformations. This brief communication presents some preliminary results on the FSI phenomena involved with the water entry of deformable wedges. Wedges with various deadrise angle and flexibility impact the water at different velocities. Two different kinds of FSI are found: (i) a repetition of impacts and separations in the fluid jet and (ii) a tendency to cavitation in the underwater fluid–structure interface. These experimental findings are important since large deformations are found to introduce clearly visible FSI phenomena commonly neglected or ignored, but crucial to correctly predict the hydrodynamic load.

1. Introduction

Hull slamming occurs when the forefoot of a ship rises above the water surface and then submerges again with high vertical velocity. Impulsive pressure loads act on the hull, introducing dynamic excitation to the structure due to the large force applied for a short period (the slamming event is in the order of milliseconds).

In the literature, many experimental works investigated the hydrodynamic pressure at the fluid–structure interface during the water entry of rigid or very stiff bodies, showing that analytical formulations like Wagner's [1] and Chuang's [2] can be used with a very high confidence. However, a severe impact might introduce large deformations that modify the fluid motion and the formulas capable of predicting the hydrodynamic impact load developed for the water entry of rigid bodies are not valid anymore. In particular, as mentioned by Korobkin et al. [3,4], the evolution of the wetted body area in time is an important characteristics of the impact, which strongly affects the magnitude of the loads. Such problems are still difficult to analyze and compute. Recently Panciroli et al. [5–7] found that the impact load during water entry of deformable wedges shows marked oscillations due to the mutual interaction between the structural deformations and the fluid flow. Predicting the structural deformations and stresses during the water entry of flexible structures is a major challenge and a deep understanding of the FSI phenomena that generate during the water entry is a dire need to correctly predict the hydrodynamic load.

The occurrence of FSI phenomena during the water entry of flexible bodies was analytically predicted in the literature (e.g. [8,9]). This work introduces some experimental findings on the FSI phenomena that generates during the water entry of flexible wedges. In particular, findings on two different kinds of FSI that never occur during the water entry of rigid or stiff structures are presented.

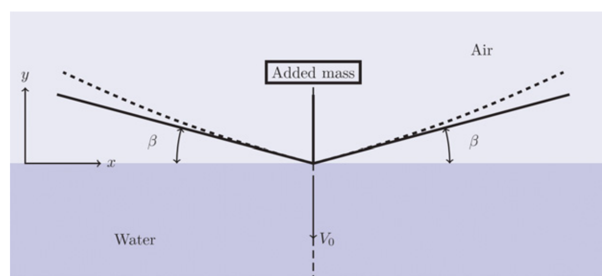

Fig. 1. Conceptual scheme of the wedge used for experiments. β = deadrise angle. Solid line: undeformed panels; dashed line: expected deformation during impact.

2. Experimental set-up

Experiments are conducted on a drop weight machine for water impacts with a maximum impact height of 4 m. Wedges are made by two panels joined together and to the falling sledge on one edge. Panels of different material and thickness can be mounted on the sledge with any deadrise angle (β in Fig. 1) ranging smoothly from 0° to 50°. Velocity, acceleration and structural deformations measurements are performed during the impact.

An high speed camera is used to capture the water entry. The capturing frequency is set to 2 kHz with a definition of 1280 × 800 pixels. A vertical transparent screen placed inside the water tank just before the wedge (clearance is ≈2 mm) prevents the out-of-plane fluid spray, which would have made it impossible to see the evolution of the fluid jet generated during the water entry. As the water on the front side of the screen will not move during the impact, the pictures show both the still water surface (on the front side of the screen) and

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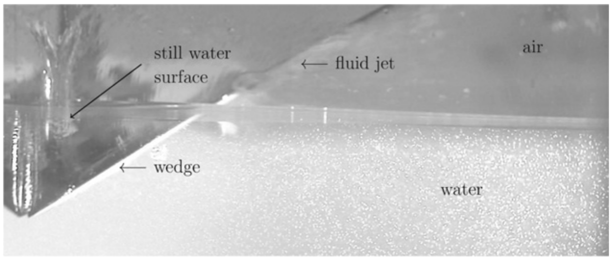

Fig. 2. Sample of image captured from the high-speed camera.

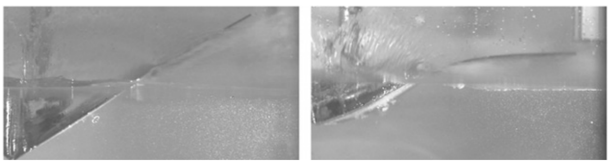

Fig. 3. Left: maximum deformation of the wedge during a "soft" impact. The wedge deformation is barely visible. Right: maximum deformation of the wedge during a "severe" impact. The free edge of the wedge is almost horizontal at its maximum deformation.

the fluid jet (on the back side of the screen), as shown in Fig. 2.

Faltinsen [10] showed that hydroelasticity is influenced by the ratio between the wetting time and the panel's lower natural frequency. To vary the fundamental natural frequency of the panels, different stiffness to area density ratios are needed. Thus, aluminum (*A*), E-glass (mat)/vinylester (*V*) and E-glass (woven)/epoxy (*W*) panels of various thickness (2 mm and 4 mm) are used. All wedges are made by two panels 300 mm long and 250 mm width. For each experimental configuration, tests are repeated three times to ensure the accuracy of the results. Repeatability of the experiments is extremely high, as the accelerations and deformations over time superpose almost perfectly and the difference between the maximum value of the recorded signals is below 5%.

3. Experimental results

The deformation of the panels varies from barely visible in case of impacts at low velocity and high deadrise angle to very large deflections in the cases of higher velocities and low deadrise angles. Two examples are shown in Fig. 3.

The next sections show that the structural deformation might introduce two different kinds of FSI phenomena that never occur during the water entry of rigid or stiff structures.

3.1. FSI – 1st kind

The water entry of stiff wedges (or flexible wedges but subjected to low impact loads) produces a water jet (with a known shape) that grows in time and progressively wets the panels. The experimental results show that lowering the deadrise angle and increasing the impact velocity might introduce important differences in the shape of the fluid jet. As an example, Fig. 4 shows the water entry of a wedge with deadrise angle of 15° entering the water at 6.7 m/s. The pictures show that there is a repetition of impacts and separations between the wedge and the fluid: in the first figure the wedge just reached the water and the fluid jet is being formed. In the second picture 1/3 of the wedge is immersed and the jet is moving at high speed to the right. In the third picture the jet is wetting the whole panel. Note that the free edge of the panel is deforming downward until now. The fourth picture shows something unknown from the literature review on the water entry of rigid bodies: the deformation of the panel is now directed upwards and the last portion of the panel (which was wet in the third picture) is now separated from the fluid. As the impact

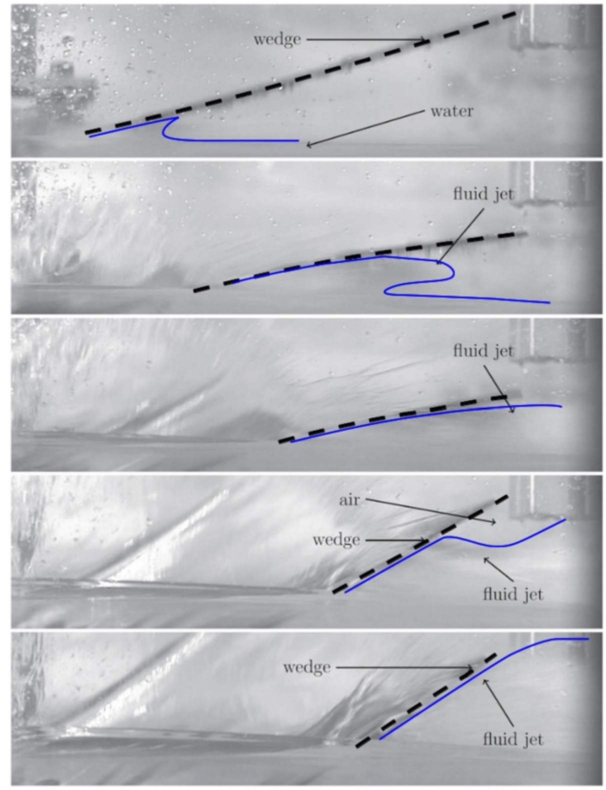

Fig. 4. Evolution of the fluid jet during the water entry of a wedge ($\beta = 15°$) at 2, 5, 7.5, 10 and 12.5 ms of impact. A fluid–structure interaction in the form of repetitions of impact and separation is visible moving from the third to the fifth picture.

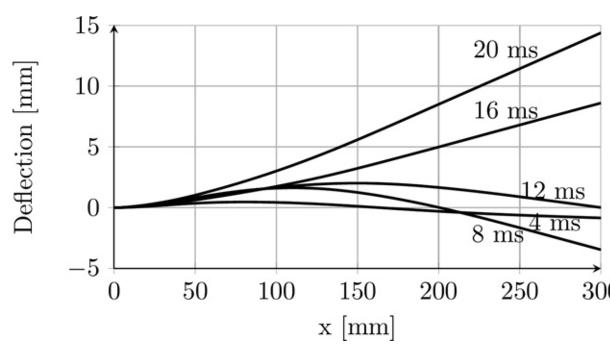

Fig. 5. Example of the wedge deflection evolution during the impact.

continues the jet will eventually wet again the whole panel, as shown in the last picture. This FSI phenomenon is due to the panel's vibrations that lead the panel to initially deform downwards, modifying the fluid jet, and later deform upwards and locally detaches from the fluid jet, as shown in Fig. 5 . A similar phenomenon was previously predicted by Korobkin and Khabakhpasheva [9] who performed an analytical parametric analysis on elastic plates.

It is thus found that the flexibility of the wedge can introduce a fluid–structure interaction phenomenon in the form of repetition of impact and separations between the structure and the fluid.

3.2. FSI – 2nd kind

The structural deformation was previously found to generate a FSI phenomenon in the fluid jet region. This section presents the findings on the FSI phenomenon generated underwater.

From the high-speed pictures it is found that the deformation

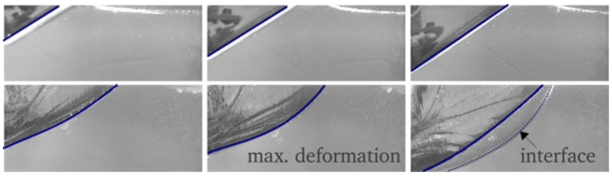

Fig. 6. Evolution in time of the water entry of a wedge. Top: "soft" impact (W, $\beta = 30^{\circ}$, $v_0 = 4.2$ m/s). Bottom: "strong" impact (V, $\beta = 20^{\circ}$, $v_0 = 6$ m/s).

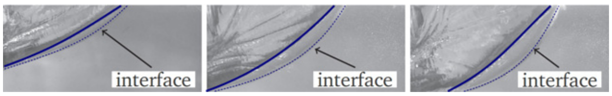

Fig. 7. Maximum dimension of the cavitation area for the water entry of a wedge (V, $\beta = 20^{\circ}$) entering the water at 4.2, 6 and 6.7 m/s.

of the panel might introduce cavitation during the water entry: as the wedge enters the water, after the maximum concave deformation is reached, a clearly visible front of cavitating fluid generates starting from the wedge underwater surface. Fig. 6 shows some examples of this phenomenon. The pictures on the first row are taken as reference; they show a wedge with deadrise angle of 30° entering the water at 4 m/s. In this case the deformation of the panel is low and the water show a smooth and uniform color. As the severity of the impact increases it is found that a cavitating area with cylindrical front generates in the fluid. The second row of pictures in Fig. 6 shows its formation: in all cases the interface of the cavitating area presents a cylindrical shape and it is generated after the maximum concave deformation is reached (central image in Fig. 6) in the panels. This phenomenon is extremely repeatable: for every repetition of the same experiment it appears with the same maximum amplitude and duration. A higher impact speed introduces higher deformations and, as the maximum concave deformation increases, so does the amplitude of the cavitating area. As an example, Fig. 7 shows the images of the maximum cavitating area generated during the water entry of a composite wedge with deadrise angle of 20° with increasing entry velocity (4.2, 6 and 6.7 m/s). The cavitating area will eventually collapse during the water entry. In the most severe impacts cases, successive cavitating pockets (with decreasing amplitude) might generate and collapse repetitively.

4. Conclusions

Large structural deformations are found to have strong effects on the fluid motion, up to generating two possible fluid–structure interaction phenomena that never occur in rigid-bodies impact: (i) the repetition of impacts and separation between the fluid and the structure in the region characterized by the fluid jet generated during the water entry and (ii) an underpressure region with a cylindrical wavefront in the underwater fluid/structure interface.

These phenomena affect the hydrodynamic pressure and the analytical formulations developed for the water entry of rigid bodies are not accurate anymore. More sophisticated tools are needed for this purpose.

These experimental findings are important since large deformations are found to introduce clearly visible FSI phenomena commonly neglected when developing new analytical formulations or numerical solutions.

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